

WTF is an LLM?

Large Language Models are the engines powering ChatGPT, Claude, Gemini, and every AI agent in the agentic economy. Here is what they actually are — explained simply, without a single equation.

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The one-liner

A Large Language Model (LLM) is a type of AI that has been trained on enormous amounts of text and learned to predict — with remarkable accuracy — what words should come next.

That sounds simple. The implications are not. From that single capability — predicting the next word — emerge the ability to write, reason, summarise, translate, code, explain, debate, and plan. It is one of the most surprising and consequential developments in the history of computing.

How does predicting words create intelligence?

This is the question that confuses most people. "Predicting the next word" sounds like autocomplete on your phone. It is not. Here is why.

To accurately predict what word comes next in a sentence, you have to understand the sentence. To understand the sentence, you have to understand the paragraph. To understand the paragraph, you have to understand the topic, the context, the nuance.

When researchers trained models on trillions of words of human text — books, articles, websites, code, scientific papers, conversations — the models did not just learn to string words together. They learned to represent meaning. They built internal models of how concepts relate to each other.

The result was not a glorified autocomplete. It was a system that could reason about problems it had never seen before, write in styles it had never been explicitly taught, and explain concepts across dozens of languages and domains.

Where did LLMs come from?

The story of LLMs is surprisingly recent and moves fast:

Year	What happened
2017	Google researchers publish "Attention Is All You Need" — the paper introducing the Transformer architecture, the technical foundation of all modern LLMs.
2018	Google releases BERT. OpenAI releases GPT-1. The race begins.
2020	OpenAI releases GPT-3 — 175 billion parameters, shockingly capable. Developers begin building on it via API.
2022	OpenAI releases ChatGPT (GPT-3.5). 100 million users in 2 months. Anthropic founded; releases Claude.
2023	GPT-4 released. Open-source models (Llama, Mistral) emerge. AutoGPT goes viral as the first autonomous agent attempt.
2024	Models capable enough for reliable agentic use. Anthropic publishes MCP. The agentic economy begins.
2025	Claude Code launches. Chinese open-weight models (DeepSeek, Qwen, Kimi) match the frontier at a fraction of the cost. Multi-hour autonomous tasks become reliable.
2026	Every major lab ships a work agent. Anthropic splits its top model into a public (Fable) and restricted (Mythos) tier. GPT-6, Gemini 3.8, Meta Muse, DeepSeek V4.

The main LLMs in 2026 — who makes what

Model names and version numbers change every few months. The labs behind them, and what each is known for, change far more slowly. Read the labs, not the numbers.

Claude (Anthropic)

The model family behind Claude Code, Cowork, Microsoft's Copilot Cowork and the default in OpenClaw. Tiers: Fable (the flagship, a new tier above Opus introduced in 2026), Opus, Sonnet and Haiku. Anthropic's Constitutional AI training makes Claude the model most agent builders reach for when reliability and honesty over long unsupervised sessions matter. Context window: one million tokens. See the [WTF is Claude](#) guide.

GPT (OpenAI)

The models behind ChatGPT, still the most widely used consumer AI product in the world, and behind OpenAI's agents: ChatGPT Work, Codex and the Agents API. GPT-6 arrived in September 2026 after the GPT-5.6 family in July. OpenAI's strength is reach: hundreds of millions of people already have the app.

Gemini (Google DeepMind)

Google's flagship family, currently Gemini 3.8, and the model behind Gemini Agent, Chrome's Auto Browse and Google's coding agent Jules. Deeply integrated with Search, Workspace and Android. Google also created the Agent2Agent (A2A) protocol for agents from different

companies to talk to each other.

Muse (Meta)

Meta's 2026 reset: a new model family and a consumer personal agent, both called Muse, delivered through WhatsApp and a standalone app with a free tier. Meta's bet is distribution — three billion people already use its messaging apps.

Grok (xAI)

Elon Musk's models, now part of SpaceX after it acquired xAI and the coding tool Cursor. The model behind Grok Bot, a team of agents sharing one cloud computer.

The Chinese open-weight models

DeepSeek, Alibaba's Qwen, Moonshot's Kimi, Zhipu's GLM and MiniMax. Released as open weights — anyone can download and run them — at a fraction of the price of US frontier models, and close enough in capability that a great many indie agents run on them. DeepSeek's cheap-to-train breakthrough in early 2025 was the moment this became a two-continent race. Guide: [WTF is Happening with China's AI](#).

Other open models

Meta's Llama family, once the default open model, has been superseded inside Meta by Muse. Mistral (France) produces efficient smaller models. OpenClaw, Paperclip and most agent frameworks let you plug in any of these by API key.

What LLMs are good at — and bad at

LLMs are good at	LLMs are bad at
Writing in any style or tone	Precise arithmetic (they approximate)
Summarising long documents	Real-time information (training data has a cutoff)
Writing and explaining code	Guaranteed factual accuracy (they can "hallucinate")
Translating between languages	Knowing what they don't know
Reasoning through multi-step problems	Consistent behaviour on identical inputs
Generating creative ideas	Tasks requiring persistent memory by default
Explaining complex topics simply	Physical world interaction (without agent tools)

Hallucination — the most important limitation

The most important thing to understand about LLMs is hallucination: the tendency to produce confident, plausible-sounding text that is factually wrong.

This happens because LLMs do not "know" facts the way a database does. They generate text based on patterns learned during training. When asked about something outside their training data, or at the edge of their knowledge, they sometimes generate text that sounds right but is not.

A lawyer once submitted a legal brief to a US court that cited cases generated by ChatGPT. The cases did not exist. The judge was not amused.

This is why human oversight of AI agents remains important — especially for high-stakes decisions. The models are getting better at knowing what they do not know, but they are not perfect. Treat LLM outputs as a very capable first draft, not a final source of truth.

Parameters, tokens, and context windows

Three technical terms come up constantly when people discuss LLMs. Here is what they actually mean:

Parameters

Parameters are the numerical values inside the model that were adjusted during training. GPT-3 had 175 billion parameters. Modern frontier models have trillions. More parameters generally means more capability — but also more cost to run. In plain English: parameters are roughly analogous to the "size" of the model's brain.

Tokens

LLMs do not process words — they process tokens. A token is roughly 3–4 characters, or about 0.75 words. "Hello, how are you?" is about 6 tokens. You pay for AI API usage in tokens — both the tokens you send (input) and the tokens the model generates (output). A typical page of text is around 500 tokens.

Context window

The context window is how much text an LLM can "see" at once — the conversation history, the document you shared, the instructions you gave. Early models had tiny context windows (4,000 tokens = about 3 pages of text). Claude's current models have a 1 million token context window — roughly 750,000 words, or the equivalent of several full novels. This is why modern agents can work with entire codebases or long documents at once.

Constitutional AI — Anthropic's approach to safety

Different companies train their LLMs differently. Anthropic's approach — Constitutional AI — is worth understanding because it directly affects how Claude behaves as an agent.

Constitutional AI is a training technique where the model is given a set of principles (a "constitution") and trained to critique and revise its own outputs against those principles. The result is a model that is more reliably helpful, honest, and harmless — not because it is restricted, but because it has internalised values.

This matters for agentic use because an agent running autonomously for hours needs to make judgment calls constantly. A model trained with Constitutional AI is more likely to handle edge cases sensibly — declining to take actions that could cause harm, flagging ambiguous situations for human review, and being honest about its limitations.

LLMs and the agentic economy

The LLM is the brain of every AI agent. Without it, an agent is just a set of if-then rules — useful but brittle. With it, the agent can:

- Understand ambiguous instructions and figure out what you actually meant.
- Handle situations it was not explicitly programmed for.
- Generate plans for achieving goals, not just execute fixed scripts.
- Read and understand unstructured data — emails, documents, web pages.
- Communicate its results in natural language.

The rapid improvement of LLMs over the past three years — from GPT-3 in 2020 to Fable, GPT-6 and Gemini 3.8 in 2026 — is the primary reason the agentic economy is happening now rather than ten years from now. The models crossed a threshold of capability that made autonomous, reliable action possible.

Glossary

LLM (Large Language Model) — An AI trained on vast amounts of text to predict and generate language. The brain inside ChatGPT, Claude, Gemini, Muse and Grok.

Transformer — The neural network architecture that underlies all modern LLMs. Introduced by Google in 2017.

Parameters — The numerical values inside an LLM adjusted during training. More parameters = generally more capable.

Token — The unit LLMs use to process text. Roughly 3–4 characters or 0.75 words. You pay for API usage in tokens.

Context window — How much text an LLM can process at once. Current frontier models: up to 1 million tokens.

Hallucination — When an LLM generates confident but factually incorrect text. A key limitation to understand.

Constitutional AI — Anthropic's technique for training models to be helpful, honest, and harmless using a set of principles.

SWE-bench — The standard benchmark for measuring LLM performance on real software engineering tasks.

Fine-tuning — Training an existing LLM further on a specific dataset to improve performance on a specific task.

Inference — Running an LLM to generate a response. The "thinking" part, as opposed to training.

Open-weight model — An LLM whose weights are publicly available — anyone can download and run it. Examples: DeepSeek, Qwen, Kimi, Mistral, Llama.

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